General S3: every characteristic, actual elliptic group

Lean proves the S3 zero-sum equivalence on the general Weierstrass model $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, over an algebraically closed field of any characteristic with nonzero discriminant. No distinct-coordinate or normal-form assumption remains. All points and sums belong to Mathlib's actual nonsingular elliptic group. [pcc:unverified]

For elementary symmetric coordinates $s_1 = x + y + z$, $s_2 = xy + xz + yz$ and $s_3 = xyz$, the checked invariant formula is

$$S_3 = s_2^2 - (4s_1 + b_2)s_3 - b_4s_2 - b_6s_1 - b_8.$$

Its equality to the source quadratic expression is checked in Lean. [pcc:unverified]

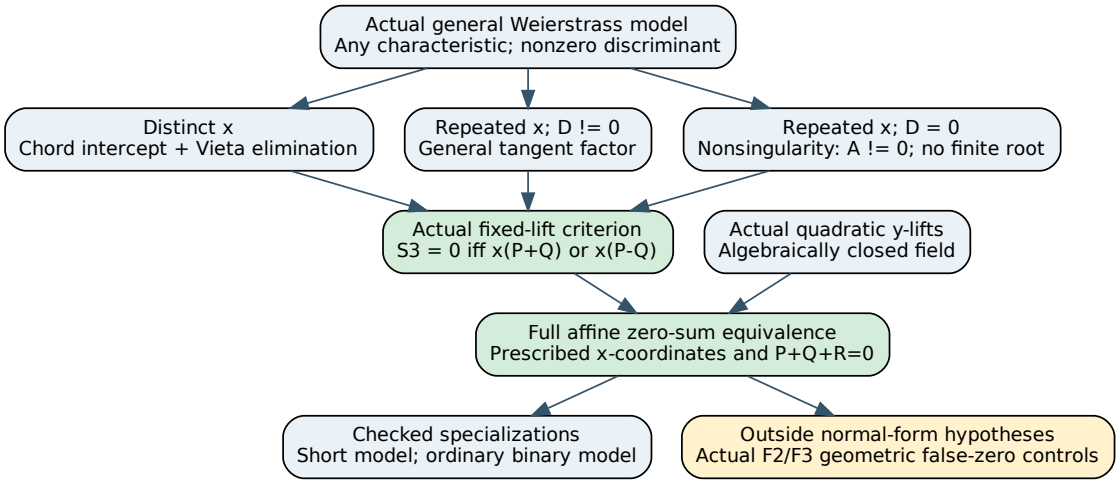

Figure 1: Actual proof dependencies. Earlier bounded native censuses are not premises. All repeated-coordinate and vertical cases are included. [pcc:unverified]

The tangent uses $A = 3x^2 + 2a_2x + a_4 - a_1y$ and $D = 2y + a_1x + a_3$. The repeated-coordinate polynomial is $A^2 + a_1AD - D^2(a_2 + 2x + z)$. If $D = 0$, nonsingularity forces $A \neq 0$; otherwise this factors through the actual doubled point. No division by 2 or 3 enters. Algebraic closure supplies actual quadratic y-coordinate lifts for the final converse. [pcc:unverified]

Whole census project gate: 167 declarations, 0 failing, 0 trusted-native exceptions. Full declaration/axiom receipt and failed-build diagnostics are retained under ../../evidence/general-s3-*. The count includes generated declarations; it is not 167 new results. Packets 4d5cd7/b32b34 remain unsigned. [pcc:unverified]

Geometric controls and source limits

Actual smooth model	Prescribed triple	Wrong-model polynomial	Actual geometric result
F2: $y^2 + y = x^3$	(1, 0, 0)	Ordinary binary expression 0	General S3 nonzero; no zero-sum triple
F3: $y^2 = x^3 + x^2 + 1$	(0, 0, 0)	Short expression 0	General S3 nonzero; no zero-sum triple

These are actual algebraic-closure point nonexistence statements, not merely missing rational-coordinate lifts. They refute dropping the model hypotheses, not either qualified formula. The normal-form specializations are themselves checked identities. Source: ../../lean/GeneralS3Controls.lean. [pcc:unverified]

Two parametric actual-group controls also compile: characteristic 2, nonsingular model with $a_1 = 0$ has trivial 2-torsion; characteristic 3, nonsingular short model has trivial 3-torsion. The latter follows by reducing the S3 diagonal to $a_4^2 \neq 0$. These statements apply over algebraically closed fields as well. They do not assume or prove the source trace/Hasse/classification or general embedding-degree bridges. [pcc:unverified]

The required rewrite certificate is IDC1h392e1740fcb7cf05. The unchanged repository native checker accepts 32 derived and 11 extra points over $2^{61}-1$; 8 variables, syntactic degree bound 5. The proof-file digest matches GeneralS3.lean. The source commit preceded issue. This is a one-host supplement; the all-characteristic/group conclusion rests on Lean, not finite samples. No new curve census or statistical inference. [pcc:unverified]

Primary review: Kosters–Yeo 2015, arXiv1503.08001v3,§2/p3, general invariant formula/Proposition2.1; page visually checked. PDF SHA256 prefix b5ea5e4c5de7. Semaev 2015, arXiv1504.01175v1,§2/pp2–3, normal-form assumptions; PDF SHA256 prefix 4db8433bdd7a. Original Semaev 2004 ePrint still 403/unread. Neither the general higher-resultant theorem nor the first-fall/complexity parts of these papers are marked audited/proved. [pcc:unverified]

Exact hashes/read scopes: ../../general-s3-sources.json. Main actual-group source: ../../lean/GeneralS3.lean, declaration generalS3_zero_iff_exists_points. Signed/independent-host standing is not asserted. Frozen earlier native sources/protocols/raw data remain unchanged; user-approved local checkpoints and zero paid spend persist. General resultant/irreducibility/degree proofs and the other 14-item obligations remain open. [pcc:unverified]